higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

CONCRETE STRUCTURES
NQF LEVEL 3

(12020343)

21 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 6 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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SECTION A

QUESTION 1: THE THEORY AND USE OF REINFORCED CONCRETE

1.1 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.1.1–1.1.10) in the ANSWER BOOK.

1.1.1 A compressive force is a pushing or a squashing force that tries to push the molecules in a material.

1.1.2 The compressive strength of concrete is its ability to carry a heavy load.

1.1.3 A tensile force is a stretching force that tries to push the structures in material further apart.

1.1.4 The tensile strength of concrete is its ability not to break when it is being pulled apart.

1.1.5 Bond failure occurs because of the aggregates used for concrete.

1.1.6 When concrete contaminates with the oil on the farm face material, it will increase the strength of the concrete.

1.1.7 Concrete starts to crack when it comes under tensile stress caused by load.

1.1.8 Corrosion is the process that helps concrete to cure.

1.1.9 Bending schedule is a detailed document that shows the side view and elevation of structures.

1.1.10 Bearing failure can occur when a bearing (or support) is too small or too weak to carry the load.

(10 × 1) (10)

1.2 Make a neat labelled drawing of a raft foundation under a vertical load. (5)

1.3 Retaining walls gives support to a slope or where the ground level has to be stepped.

Make a neat labelled sketch of a labelled cantilever retaining wall. (5)

1.4 Make a neat labelled drawing of a pile foundation under a vertical load. (5)

[25]

QUESTION 2: REINFORCED CONCRETE IN CONSTRUCTION

2.1 Choose a/an item/word/description from COLUMN B that matches a/an item/word/description in COLUMN A. Write only the letter (A–F) next to the question number (2.1.1–2.1.5) in in the ANSWER BOOK.

COLUMN A		COLUMN B	
2.1.1	A process of increasing the strength of concrete by installing steel bars or wire mesh or strands of glass reinforced plastic in the concrete or before casting concrete	A	cold rolling
		B	reinforced concrete
		C	high-yield steel
		D	mild steel
2.1.2	Steel that has high carbon content and high strength, but is very brittle	E	cold working
2.1.3	Steel made up of iron with less than 30% carbon and it can be easily cut, bent or welded	F	sheet metal drawing
2.1.4	Changing the shape and size of a metal at room temperature		
2.1.5	Metalwork process in which metal is changed by passing it through rollers at a low temperature		

(5 × 1) (5)

2.2 Name and describe TWO types of cold drawings. (4)

2.3 List and explain FOUR different symbols used to designate different types of reinforced steel. (8)

2.4 List THREE commonly used sizes/nominal diameters of reinforcement. (3)

2.5 Illustrate the problem and solution by making a neat labelled drawing of a ground slab with a contraction joint. The mesh is in the bottom half of the slab. (5)

[25]

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QUESTION 3: REINFORCING DOCUMENTATION

3.1 Explain the following terms:

3.1.1 Bending schedule
3.1.2 Prefix
3.1.3 Tolerance
3.1.4 Convention
3.1.5 Suffix

(5 × 1) (5)

3.2 Explain the following different types of lines used on the reinforcing drawing:

3.2.1 Continuous thick lines

3.2.2 Continuous thin lines

3.2.3 Broken lines

3.2.4 Chain dotted lines

3.2.5 Shaded portions or hatched areas

(5 × 1) (5)

3.3 Use the information given in FIGURE 1 below and make TWO neat cross-sectional drawings of the concrete beam at AA and BB to show the hidden bars and detail in FIGURE 1. The drawings must include the placing of the beams. All the details and labels must be included.

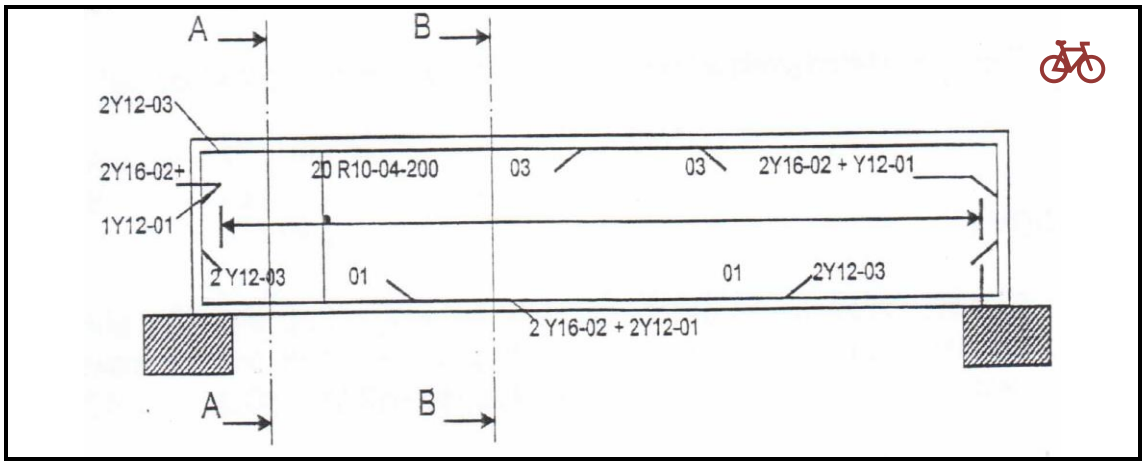


FIGURE 1

(15)
[25]

QUESTION 4: FABRICATION OF REINFORCING STEEL

4.1 Write down the EIGHT steps a steel bender should follow when he/she has to prepare and bend steel for a client. (8)

4.2 Make TWO neat drawings to show the TWO methods of splicing recommended by SANS 10144. (7)

4.3 Give THREE reasons why steel must be securely installed to stay in place. (3)

4.4 You have to fix the reinforcement for a concrete beam.

Explain in FIVE steps the order in which reinforced bars for beams are fixed. (5)

4.5 What is the difference between the following terms?

4.5.1 A reinforced cage

4.5.2 A mat

(2 × 1) (2)

[25]

TOTAL: 100

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